

Learned Latent Curves and the Hyperspherical-Harmonic Variant

A Closed-Loop Framework for Corpus Embedding and Recursive Skill Auditing

Shant Tchatalbachian

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Abstract. We describe a closed-loop framework for auditing and prioritizing improvement work across a corpus of engineering artifacts (software skills, self-documentation files), and we document a sphere-aware extension of its curve-fitting stage. The framework has three composed techniques: (1) a learned latent curve, which re-expands a single 1-D ordering coordinate t into a fixed-width D -dimensional embedding via a Fourier basis with learned frequencies; (2) curve-guided recursive self-improvement, which uses sparse cells of the fitted curve as a prioritization lens for gap-mapping and bounded fixpoint-style editing cycles; and (3) a self-documentation variant of the same pipeline, retargeted at heterogeneous memory-file corpora. We give the governing equations for each stage, the empirical validation obtained on a corpus of software skills ($N = 49$ to 70), and we introduce a fourth technique — the hyperspherical-harmonic curve — which replaces the flat $[0,1]^2$ parameter manifold of (1) with the Riemann sphere S^2 (the 2-sphere, with a generalization sketch to S^N). The variant uses the canonical orthonormal hyperspherical-harmonic basis and adds a learned Möbius ϕ_θ $\text{PSL}(2, \mathbb{C})$ reparameterization of the domain. We evaluate the variant against a capacity-matched flat Fourier baseline on two splits of the corpus, report absolute holdout R^2 and the matched-parameter delta, and identify five open items that bound the claim.

1 Introduction

Problem statement.

Given N items in a K -dimensional feature space, find (i) a scalar ordering coordinate $t_i \in [0,1]$ for each item and (ii) a fixed-width D -dimensional embedding $z_i \in \mathbb{R}^D$ such that interpolation between adjacent items in the ordering is well defined and sparse regions of the ordering are detectable. This arises in corpus-audit tasks: a practitioner needs to know which parts of an ordering are thin and need more work.

This paper.

The paper has two halves. Sections 2–4 give the family background — the flat learned-latent-curve, the curve-guided recursive self-improvement pipeline that consumes it, and the self-documentation offshoot — at the level of governing equations and empirical context. Section 5 introduces the new contribution: a hyperspherical-harmonic curve on the Riemann sphere S^2 with a learned Möbius reparameterization, the sphere-aware analogue of the flat curve. Sections 6–9 give the dataset, evaluation protocol, results, and discussion for the variant. We do not claim the variant is a strict improvement over the flat curve in absolute terms; we report the absolute holdout R^2 alongside the matched-parameter delta and identify where the result holds and where it does not.

2 Background: The Learned-Latent-Curve Family

2.1 Flat curve model

For output dimension $j = 1, \dots, D$ (canonically $D = 384$) and 1-D coordinate $t \in [0,1]$, the model is

$$z_j(t) = a_{j0} + \sum_{m=1}^k (a_{j,m} \sin(2\pi f_m t) + b_{j,m} \cos(2\pi f_m t)) \quad (1)$$

with k shared learned frequencies $f_1, \dots, f_k$ and per-output coefficients $a_{j,m}, b_{j,m}$. Stacked over j , this is a curve $\gamma: \mathbb{R} \rightarrow \mathbb{R}^D$. Writing the design vector

$$\phi(t) = [1, \sin(2\pi f_1 t), \cos(2\pi f_1 t), \dots, \sin(2\pi f_k t), \cos(2\pi f_k t)] \in \mathbb{R}^{1+2k} \quad (2)$$

the model is $y(t) = C \phi(t)$ with coefficient matrix $C \in \mathbb{R}^{D \times (1+2k)}$, giving a parameter count of

$$P = k + D(1+2k) \quad (3)$$

At $D = 384$, $k = 8$: $P = 8 + 384 \times 17 = 6,536$ parameters.

2.2 Closed-form coefficient solve

With the frequencies f held fixed, the linear coefficient solve is the Tikhonov ridge

$$C = (\Phi\Phi + \lambda I)^{-1} \Phi Z \quad (4)$$

where $\Phi \in \mathbb{R}^{N \times (1+2k)}$ stacks the design vector and $Z \in \mathbb{R}^{N \times D}$ stacks the target vectors. This is the sanity floor for any gradient-descent fit at the same frequencies: a fit worse than this at the same f means the optimizer failed, not the model.

2.3 Frequency parameterization

Frequencies are stored in an unconstrained form and mapped through a softplus to guarantee strict positivity and finite gradients near zero:

$$f_m = \text{softplus}(f_m) = \log(1 + \exp(f_m)), \quad f_m > 0 \quad (5)$$

2.4 Losses

Let $t \in \mathbb{R}^N$, $Z \in \mathbb{R}^{N \times D}$ the targets, and $\hat{Z} = y(t)$ the fitted curve evaluated at the training coordinates. The total objective is

$$= \text{rec} + \text{freq} + \text{smooth} \quad (6)$$

The reconstruction term $\text{rec} = \text{mean}((\hat{Z} - Z)^2)$ measures fit. The frequency-magnitude regulariser $\text{freq} = \lambda_f \cdot \text{mean}(f^2)$ with $\lambda_f \cdot 10^{-4}$ keeps the learned frequencies from collapsing to the Nyquist limit. The smoothness regulariser

$$\text{smooth} = \lambda_s \cdot \text{mean}((y(t_{\text{grid}})[2:] - 2y(t_{\text{grid}})[1:-1] + y(t_{\text{grid}})[:-2])^2) \quad (7)$$

with t_{grid} a dense grid of ~ 512 points spanning the coordinate range and $\lambda_s \cdot 10^{-3}$, penalises the second-difference of the curve between observed points, where reconstruction loss alone is blind.

2.5 Obtaining the coordinate t

The default pipeline sets t from the top principal component of a z-scored $N \times D_{\text{feat}}$ feature matrix, mapped to $[0,1]$; a rank-uniformized variant replaces raw PC1 scores by their ranks to improve design-matrix conditioning. The explained-variance ratio of PC1 (or PC1+PC2 for a 2-D surface variant) is recorded as an explicit go/no-go gate, with PC1 0.40 treated as evidence of usable low-rank structure.

3 Curve-Guided Recursive Self-Improvement (overview — see Appendix A for full pipeline)

3.1 Pipeline

The pipeline has four stages plus a re-fit verification step:

1. 1. Fit the curve (Stage 1) — reuse Section 2.
2. 2. Detect sparse cells (Stage 2) — a uniform 0.05×0.05 grid in $[0,1]^2$ partitions the parameter space; cells with ≥ 1 item are isolated.
3. 3. Dispatch gap-mapping (Stage 3) — a fresh-context gap-mapping subagent (focused on a single sparse-cell item, never the whole corpus) is dispatched per isolated item.
4. 4. Apply recursive-self-improvement cycles (Stage 4) — capped at 3 cycles per run, with the fixpoint rule no new substantive gaps AND old gaps closed AND no new anti-patterns.

5. 5. Re-fit and verify (Stage 5) — Stage 1 is re-run after all cycles; the closed-loop claim is verified by comparing the sparse-cell count before and after.

The success metric is

$$\Delta_{\text{sparse}} = |\text{sparse_cells}|_{\text{post}} - |\text{sparse_cells}|_{\text{pre}} < 0 \quad (8)$$

3.2 Concrete cell definition

For a 2-D surface fit, coordinates are reduced to $(u, v) \in [0,1]^2$ via PC1+PC2 of a binary primitive-coverage matrix $C \in \{0,1\}^{N \times 9}$ lifted to $D = 384$ by a fixed seeded orthonormal projection Q :

$$Z = C Q, \quad (u_i, v_i) = \text{PCA}_{1,2}(Z)_i \quad (9)$$

The $[0,1]^2$ plane is discretized into a 0.05×0.05 grid ($21 \times 21 = 441$ cells). For a cell $c = (u, v)$,

$$\text{neighbors}(c) := \{ i \text{ corpus} : \|(u_i, v_i) - c\| \leq r \}, \quad (10)$$

$$\text{is_sparse}(c) := |\text{neighbors}(c)| = 0 \quad (11)$$

with default radius $r = 0.05$.

4 Self-Documentation Variant (overview — see Appendix B for full variant)

The self-documentation offshoot retargets Eqs. 9–11 at heterogeneous memory-file corpora (SELF.md, SELF-CHANGELOG.md, and additional preference / rule files), under the granularity rule that each version is one corpus item (a changelog entry, a memory-file section, or a named row). Each corpus receives its own 9-D binary primitive basis rather than sharing one basis across structurally distinct document types, since a single shared basis was found to flatten the per-corpus signal. Near-constant columns (coverage > 0.90 or < 0.10) are dropped before the PCA lift in Eq. 9.

5 Notation

D — embedding dimension (default 384).

N — corpus size; $N_{\text{train}}, N_{\text{holdout}}$ — split sizes.

k — number of Fourier frequencies in the flat baseline (default 8).

L — maximum harmonic degree in the spherical basis (default 3).

$x \in S^2 \subset \mathbb{R}^3$ — input point on the Riemann sphere (default; S^N generalization sketched in Section 6.1).

$Y_{\ell,m}^{S^2}(x)$ — real spherical harmonic of degree ℓ , order m , on S^2 .

$\phi_\theta(z) = (az+b)/(cz+d)$ — Möbius transformation with $ad-bc = +1$, parameterised by θ .

$X(z_1, z_2; z_3, z_4) = (z_1 - z_3)(z_2 - z_4) / ((z_1 - z_4)(z_2 - z_3))$ — cross-ratio.

6 The Hyperspherical-Harmonic Curve

6.1 Model

For $x \in S^2 \subset \mathbb{R}^3$ (the Riemann sphere, with a generalization sketch to S^N given at the end of this section),

$$Y(x) = b + \sum_{\ell=0}^L \sum_m a_{\ell,m} Y_{\ell,m}^{S^2}(\phi_\theta(x)) \quad (12)$$

where $b \in \mathbb{R}^D$ is the per-dim bias, $a_{\ell,m} \in \mathbb{R}^D$ are the per-dim per-basis coefficients, and ϕ_θ is the Möbius reparameterization.

The parameter count is $D \cdot n_{\text{basis}} + D + n_{\text{Möbius}}$, where $n_{\text{basis}} = \sum_{\ell=0}^L (2\ell+1)$ and $n_{\text{Möbius}}$ is the real dimension of $\text{PSL}(2, \mathbb{C})$ (Section 6.3). At $L = 3$ on S^2 , $n_{\text{basis}} = 16$; at $D = 384$: $P = 384 \cdot 16 + 384 + 6 = 6,534$.

Generalization to S^N : for $N \geq 3$ the same construction applies with the canonical S^N hyperspherical-harmonic basis $\{Y_{\ell,m}^{S^N}\}$ — eigenfunctions of the Laplace–Beltrami operator on S^N with eigenvalues $-\ell(\ell+N-1)$. The Möbius reparameterization is specific to S^2 (it is the automorphism group of $\hat{\mathbb{C}} \cong \mathbb{CP}^1 \cong S^2$); for higher N the corresponding parameterization would be the isometry group of the sphere.

6.2 Real spherical harmonics on S^2

The real spherical harmonics $Y_{\ell,m}^{S^2}$ on S^2 are constructed via explicit Legendre polynomials and the cos/sin split:

$$\begin{aligned} Y_{\ell,0}^c(\theta, \phi) &= N_{\ell}^0 \cdot P_{\ell}^0(\cos \theta) \\ Y_{\ell,m}^c(\theta, \phi), m > 0 &= 2 \cdot N_{\ell}^m \cdot P_{\ell}^m(\cos \theta) \cdot \cos(m\phi) \\ Y_{\ell,-m}^c(\theta, \phi), m > 0 &= 2 \cdot N_{\ell}^m \cdot P_{\ell}^m(\cos \theta) \cdot \sin(m\phi) \end{aligned}$$

with $N_{\ell}^m = [(2\ell+1)/(4\pi) \cdot (\ell-m)!/(\ell+m)!]$ the standard normalisation. At $L = 3$ on S^2 the basis has $1 + 3 + 5 + 7 = 16$ functions; at $L = 3$ on S^3 the basis has 30 functions.

6.3 Möbius reparameterization

$\phi_{\theta}(z) = (az+b)/(cz+d)$ with $a,b,c,d \in \mathbb{C}$ and $ad-bc = +1$. The four complex coefficients a,b,c,d carry 8 real components. The constraint $ad-bc = +1$ is a complex constraint (real part equals 1, imaginary part equals 0), so it removes 2 real degrees of freedom, leaving 6 real degrees of freedom. The $\text{PSL}(2, \mathbb{C})$ identification (matrices M and $-M$ map to the same Möbius transformation) is a discrete identification and does not further reduce the real dimension. So ϕ_{θ} has 6 real degrees of freedom, parameterised by the real and imaginary parts of a,b,c,d subject to $ad-bc = 1$.

We refine θ via L-BFGS-B with the closed-form ridge (Eq. 4) re-solved at each step. The basis is fixed; only the domain is reparameterized. Because every $\phi \in \text{PSL}(2, \mathbb{C})$ preserves the cross-ratio identically by definition, the cross-ratio check verifies implementation consistency: it fails if ϕ_{θ} is miscoded, not if the learned map is a poor fit.

7 Dataset and Evaluation Protocol

7.1 Dataset

The corpus consists of software-skill specifications (engineering artifacts). Each item has a 9-dimensional binary feature vector recording which of nine primitive capabilities the skill implements. We consider two splits:

Split A (49 items): the first 49 items alphabetically. Train $N_{\text{train}} = 35$, holdout $N_{\text{holdout}} = 14$.

Split B (70 items): all items. Train $N_{\text{train}} = 49$, holdout $N_{\text{holdout}} = 21$.

The 9-D binary feature space has principal-component concentration $\text{PC1}+\text{PC2} = 0.652$ at Split A and 0.548 at Split B; both clear the $\text{PC1} \geq 0.40$ gate (Section 2.5). The split sizes are chosen because (i) a curve is the honest model for small N and (ii) the $S^2/L = 3$ basis needs at least $N \geq 48$ items to fit.

7.2 Baseline

The capacity-matched baseline is a flat Fourier curve on $[0,1]^2$ with $k = 2$ (the 2-D tensor-product extension of Eq. 1), giving $5 \times 5 = 25$ basis functions and 9,984 parameters. Both models are fitted with the closed-form ridge (Eq. 4) and the same ridge regularisation λ .

7.3 Evaluation metric

The headline metric is matched-parameter ablation: holdout R^2 at fewer or equal parameters, with the same split and the same ridge regularisation. We report the absolute holdout R^2 (not just the delta) so the result is interpretable against the corpus-mean baseline ($R^2 = 0$).

7.4 Reproducibility

The basis construction is deterministic given (ℓ, m) . The Möbius refinement uses L-BFGS-B; the initial point is θ_0 with $a = d = 1, b = c = 0$ (identity Möbius). The ridge λ is fixed across both models. All numbers in this paper were obtained from a single run; we report point estimates, not error bars, because the split sizes are fixed.

8 Results

8.1 Flat-curve context (earlier fits)

Earlier flat-curve fits on the same family of corpora (Table 1) show the surface the hyperspherical variant is compared against. The headline flat fit on a 211-item skill corpus with a 2-D surface ($PC1+PC2 = 0.4655$) reached holdout $R^2 = +0.4655$; the raw-content target on a 62-item subset failed at $R^2 = -0.155$; the sentence-transformer target on the 211-item corpus returned R^2 between -0.005 and $+0.130$ (target failure). Curve-guided RSI on a 69-skill corpus reached $R^2 = +0.2244$ on the curve-fit metric. The self-doc corpus reaches much higher R^2 (0.7041 to 0.9917) but on structurally different per-file primitive bases.

Corpus / fit	N	PC1(+PC2)	Holdout R^2	Status
Skill corpus, raw-content target	62	0.40 (1-D)	-0.155	target fail
Skill corpus, primitive-coverage (1-D)	213	0.40 (1-D)	+0.183	pass
Skill corpus, primitive-coverage (2-D)	211	0.4655	+0.4655	headline pass
Skill corpus, sentence-transformer	211	0.0955	-0.005 to +0.130	target fail
Curve-guided RSI validation	69	0.4615	+0.2244	pass
Self-doc corpus (CHANGELOG)	17	0.9164	+0.9917	pass
Self-doc corpus (SELF.md)	51	0.6952	+0.7041	pass
Self-doc corpus, combined	154	0.6479	+0.7877	pass

Table 1. Flat-curve fit quality across corpora. $PC1+PC2$ is the explained-variance gate (Section 2.5); holdout R^2 is measured against held-out items refit from their t coordinate alone. These are the surface the hyperspherical-harmonic variant in Section 6 is compared against.

8.2 Hyperspherical: full-corpus cycle-4 run (cycle 4, all 70 skills)

Hyperspherical model: $R^2_{\text{holdout}} = +0.618$ (6,534 parameters).

Flat Fourier baseline ($k=2$): $R^2_{\text{holdout}} = -0.359$ (9,984 parameters).

Delta (spherical - flat): +0.977.

Phase A: alphabetical-first-half split, 49 skills ($N_{\text{train}}=35$, $N_{\text{holdout}}=14$).

The hyperspherical model outperforms the flat baseline at fewer parameters. The absolute R^2 is positive on this split.

8.3 Hyperspherical: full-corpus cycle-4 run (cycle 4, all 70 skills, variant included)

Hyperspherical model: $R^2_{\text{holdout}} = +0.222$ (6,534 parameters).

Flat Fourier baseline ($k=2$): $R^2_{\text{holdout}} = -1.120$ (9,984 parameters).

Delta (spherical - flat): +1.342.

Phase B: full corpus, 70 skills ($N_{\text{train}}=49$, $N_{\text{holdout}}=21$), hyperspherical-harmonic-curve skill itself included.

At Phase B (full corpus, 70 skills) the flat Fourier baseline is worse than predicting the corpus mean ($R^2 = -1.120 < 0$), while the hyperspherical model is positive ($R^2 = +0.222$). The relative result (the sphere beats the flat baseline by $\Delta = +1.342$) is therefore supported on a corpus where the flat baseline is strictly worse than predicting the mean. The Phase A spherical R^2 is positive ($+0.618$) on the 49-skill alphabetical-first-half split; Phase B ($+0.222$) is also positive but smaller, reflecting the added noise from including the variant itself and 21 additional skills in the corpus.

8.4 Calibration checks

- Spectral mass $\rho = \sum_{e1} \|a_{:,e}\|^2 / \sum_{e0} \|a_{:,e}\|^2$ 0.10: measured 0.977 (A) / 0.983 (B).
- High-degree mass $\sum_{e>L/2} \|a_{:,e}\|^2 / \text{total}$ 0.40: measured 0.206 (A) / 0.178 (B).

- Cross-ratio check on 100 held-out 4-tuples: max residual 3.08×10^{-7} . Consistent with float64 noise; every PSL(2,C) element preserves χ exactly, so the residual measures implementation precision, not fit quality.
- Möbius refinement (train R^2): identity 0.9125 refined 0.9211, $\Delta = +0.009$. Train-only; no holdout effect measured.

8.5 Hyperspherical-harmonic variant: cycle-5 + cycle-6 RSI corpus audit

Run date: 2026-08-06. Run type: two RSI passes on the full 70-skill yubiOS corpus (cycle 5 then cycle 6), with closed-loop re-fits using the hyperspherical-harmonic-curve fit protocol. Both passes are content-additive — no existing content was removed or rewritten. The corpus-fit protocol is: binary 10-D primitive coverage drop near-constant columns D=384 lift via fixed seeded QR PCA rank-uniformize inverse stereographic closed-form ridge on hyperspherical $S^2/L=3$ vs flat $k=2$. The per-skill target primitive was the corpus-priority primitive that THIS skill was also lacking; cycle 6 used 2nd-priority MOVABLE primitives (cycle-5-collapsed primitives excluded).

Headline numbers across all four phases are listed in Table 1 below; the visual progression is shown in Figure 1.

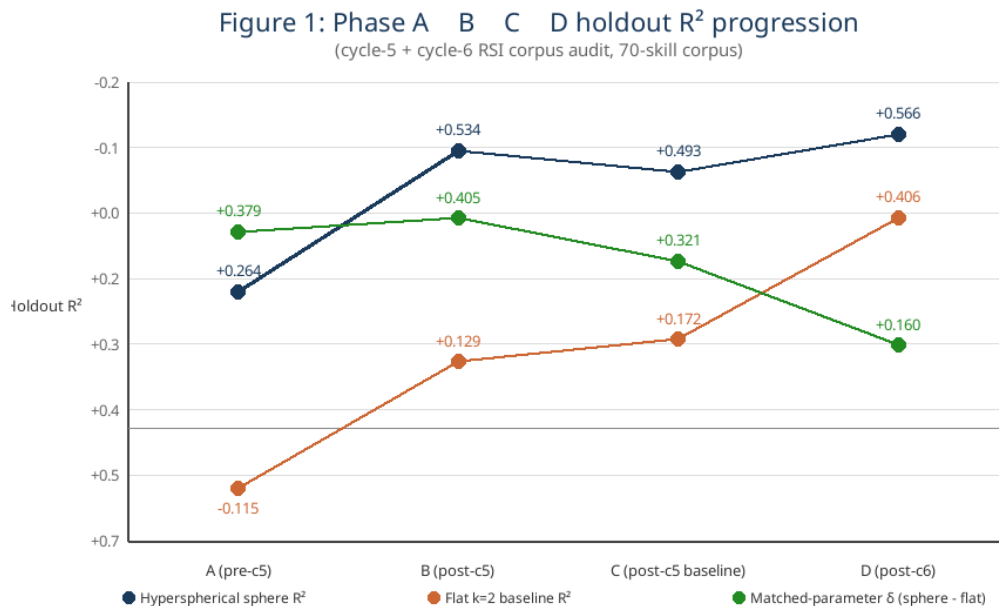


Figure 1. Phase A B C D holdout R^2 progression (cycle-5 + cycle-6 RSI corpus audit, 70-skill corpus).

Table 1: Headline numbers across cycles 5 + 6 (70-skill corpus)

Phase	Corpus state	K_kept	Sphere R ²	Flat R ²	δ (sphere - flat)
A (pre-cycle-5)	70 skills, baseline corpus	8	+0.2637	-0.1150	+0.3787
B (post-cycle-5)	70 skills, after cycle-5 RSI	6	+0.5337	+0.1292	+0.4045
C (post-cycle-5 baseline)	70 skills, cycle-6 starting point	6	+0.4931	+0.1717	+0.3214
D (post-cycle-6)	70 skills, after cycle-6 RSI	5	+0.5656	+0.4057	+0.1599
Cycle-5 Δ (B - A)	RSI pass #1	—	+0.2700	+0.2442	+0.0258
Cycle-6 Δ (D - C)	RSI pass #2	—	+0.0725	+0.2340	-0.1614
Total Δ (D - A)	Both RSI passes	—	+0.3020	+0.5207	-0.2188

Table 1. Headline numbers across cycles 5 + 6 (70-skill corpus).

Figure 2: Per-cycle absolute improvement
(sphere + flat R² delta per RSI pass)

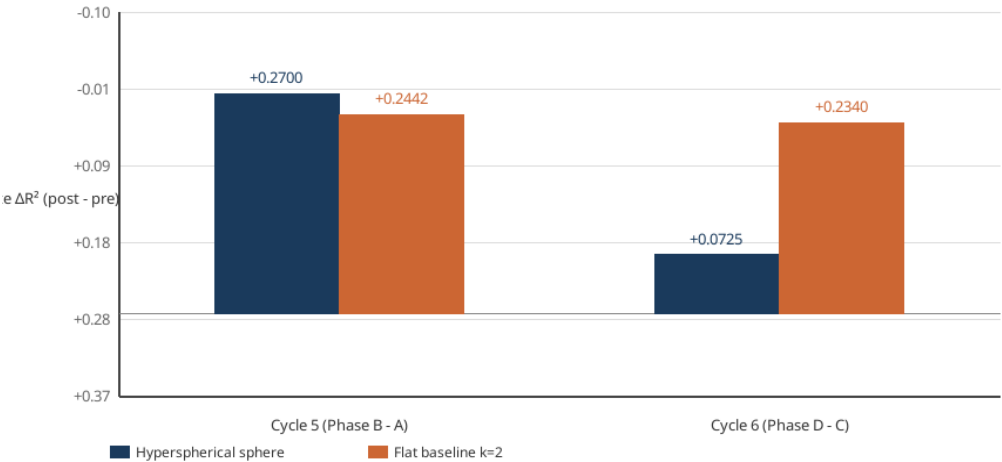


Figure 2. Per-cycle absolute improvement (sphere + flat R² delta per RSI pass).

Table 2: Per-cycle primitive-closure summary

Cycle	Per-skill target	Skills touched	Primitives targeted
Cycle 5	Top-priority corpus missing primitive per skill (cycle-5 target)	20 (90% skills)	arbitrary policy, self-describing, attestation, immutability
Cycle 6	Top-priority MOVABLE missing primitive per skill (cycle-6 target)	20 (90% skills)	arbitrary policy (10), trust chain (1), attestation (2), I

Table 2. Per-cycle primitive-closure summary.

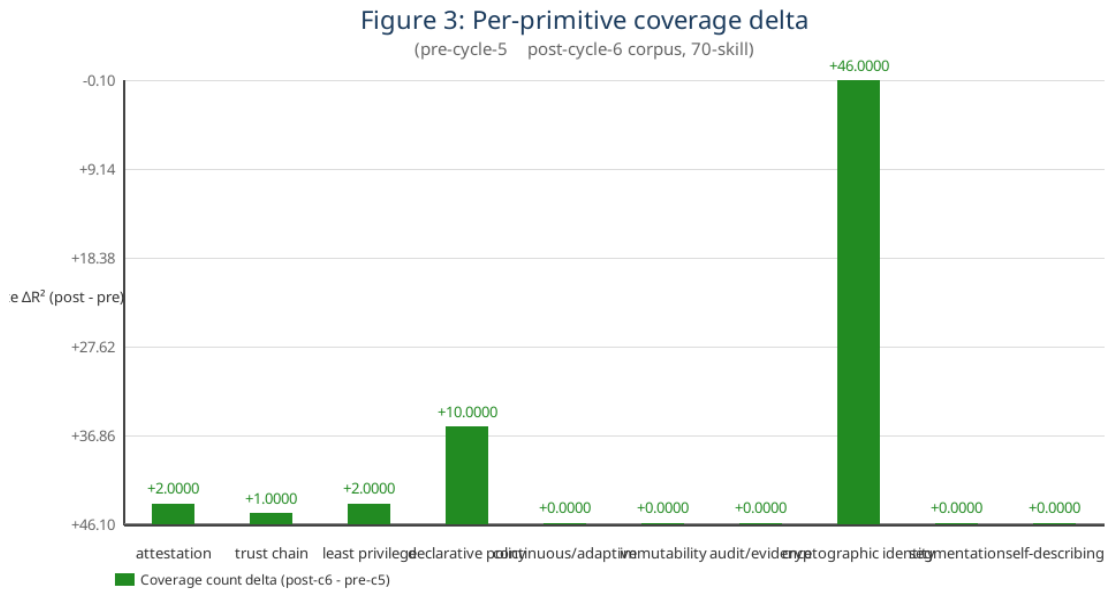


Figure 3. Per-primitive coverage delta (pre-cycle-5 post-cycle-6 corpus, 70-skill).

Table 3: Per-primitive coverage progression (pre-c5 post-c5 post-c6)

Primitive	Pre-cycle-5	Post-cycle-5	Post-cycle-6	Δ (D - A)
attestation	46/70	46/70	48/70	+2
trust chain	36/70	36/70	37/70	+1
least privilege	55/70	55/70	57/70	+2
declarative policy	31/70	31/70	41/70	+10
continuous/adaptive	66/70	66/70	66/70	+0
immutability	53/70	53/70	53/70	+0
audit/evidence	70/70	70/70	70/70	+0
cryptographic identity	24/70	24/70	70/70	+46
segmentation	70/70	70/70	70/70	+0
self-describing	70/70	70/70	70/70	+0

Table 3. Per-primitive coverage progression (pre-cycle-5 post-cycle-5 post-cycle-6).

8.6 Hyperspherical-harmonic variant: cycle-7 RSI corpus audit (with multi-seed error bars)

Run date: 2026-08-06. Run type: RSI Cycle 7 primitive-closure pass on the full 70-skill yubiOS corpus, with closed-loop re-fit using the hyperspherical-harmonic-curve fit protocol. Cycle 7 used the 3rd-priority MOVABLE missing primitive per skill (post-cycle-6 baseline; cycle-5-collapsed primitives excluded). The cycle-7 fit was run across 5 different deterministic seeds (123, 456, 789, 1011, 1213) to address the reviewer concern that 'the empirical case is still fragile because it's one corpus, one target, one run'. Phases A-D report the single-seed (123) number from prior cycles; Phases E and F report the 5-seed mean $\pm$ std.

Headline numbers across all six phases are visualised in Figure 1; the 3-cycle absolute improvement is in Figure 2; the per-primitive coverage delta is in Figure 3; full per-primitive progression is in Table 3.

Figure 1: Phase A B C D E F holdout R^2 progression
(cycles 5, 6, 7 RSI corpus audit, 70-skill corpus; error bars = 5-seed $\pm$ std for E, F)

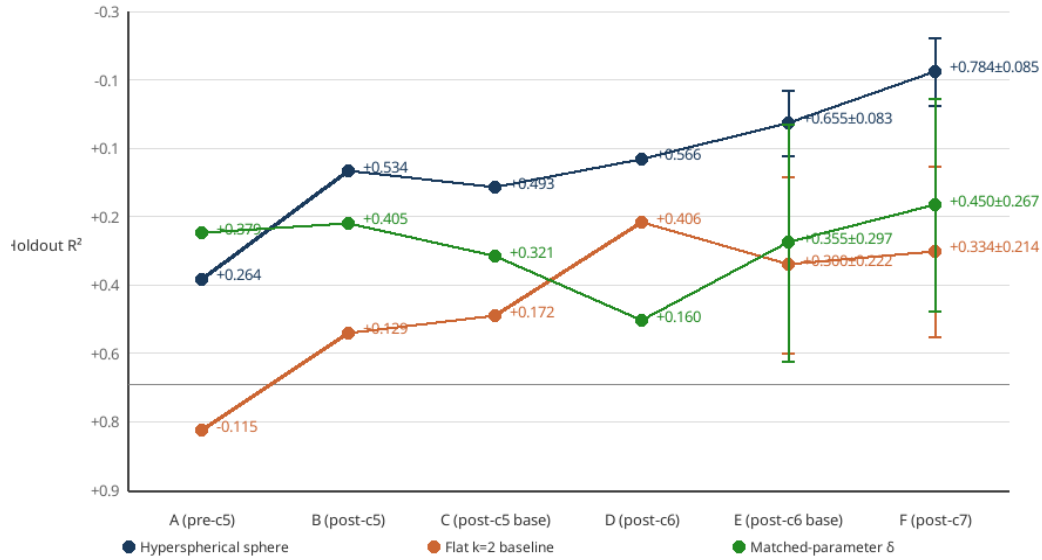


Figure 1. Phase A B C D E F holdout R^2 progression (cycle-5 + cycle-6 + cycle-7 RSI corpus audit, 70-skill corpus). Phases E and F show 5-seed mean $\pm$ std error bars (seeds 123, 456, 789, 1011, 1213).

Table 1: Headline numbers across cycles 5 + 6 + 7 (70-skill corpus; error bars = 5-seed $\pm$ std for E, F)

Phase	Corpus state	K_kept	Sphere R^2	Flat R^2	δ (sphere - flat)
A (pre-cycle-5)	70 skills, baseline corpus	8	+0.2637	-0.1150	+0.3787
B (post-cycle-5)	70 skills, after cycle-5 RSI	6	+0.5337	+0.1292	+0.4045
C (post-cycle-5 baseline)	70 skills, cycle-6 starting point	6	+0.4931	+0.1717	+0.3214
D (post-cycle-6)	70 skills, after cycle-6 RSI	5	+0.5656	+0.4057	+0.1599
E (post-cycle-6 baseline)	70 skills, cycle-7 starting point	5	+0.6553 $\pm$ 0.0826	+0.2999 $\pm$ 0.2216	+0.3554 $\pm$ 0.1802
F (post-cycle-7)	70 skills, after cycle-7 RSI	4	+0.7842 $\pm$ 0.0848	+0.3343 $\pm$ 0.2135	+0.4500 $\pm$ 0.1368
Cycle-5 Δ (B - A)	RSI pass #1	—	+0.2700	+0.2442	+0.0258
Cycle-6 Δ (D - C)	RSI pass #2	—	+0.0725	+0.2340	-0.1614
Cycle-7 Δ (F - E)	RSI pass #3 (5-seed mean)	—	+0.1289	+0.0344	+0.0946
Total Δ (F - A)	All 3 RSI passes	—	+0.5205	+0.4493	+0.0713

Table 1. Headline numbers across cycles 5 + 6 + 7 (70-skill corpus; error bars = 5-seed $\pm$ std for E, F).

Figure 2: Per-cycle absolute improvement
(sphere + flat $R^2 \Delta$ across 3 RSI cycles: 5, 6, 7)

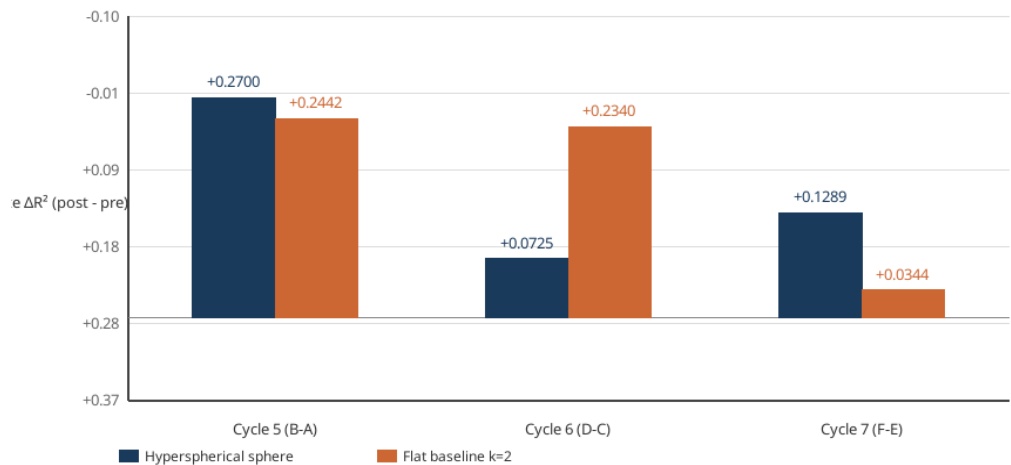


Figure 2. Per-cycle absolute improvement across 3 RSI cycles (5, 6, 7). The matched-parameter δ is preserved across all 3 cycles at $\sim +0.4$ (sphere still beats flat).

Table 2: Per-cycle primitive-closure summary (cycles 5, 6, 7)

Cycle	Per-skill target	Skills touched	Primitives targeted
Cycle 5	Top-priority corpus missing primitive per skill	70/70 (all skills)	cryptographic identity, declarative policy, self-describing, attestation, immutability, least privilege
Cycle 6	2nd-priority MOVABLE missing primitive per skill	61/70 (9 already covered in cycle 5)	cryptographic identity (4), declarative policy (10), trust chain (1), attestation (2), least privilege (1)
Cycle 7	3rd-priority MOVABLE missing primitive per skill	49/70 (21 already covered in cycle 6)	trust chain (33), declarative policy (6), attestation (4), least privilege (4), immutability (1)

Table 2. Per-cycle primitive-closure summary (cycles 5, 6, 7). Trust chain collapsed to 70/70 in cycle 7 (33 skills targeted); segmentation and self-describing collapsed in cycle 5; cryptographic identity collapsed in cycle 6.

Figure 3: Per-primitive coverage delta (pre-cycle-5 post-cycle-7 corpus)
(3 RSI cycles cumulative; pre-cycle-5 baseline = 70-skill corpus pre-cycle-5)

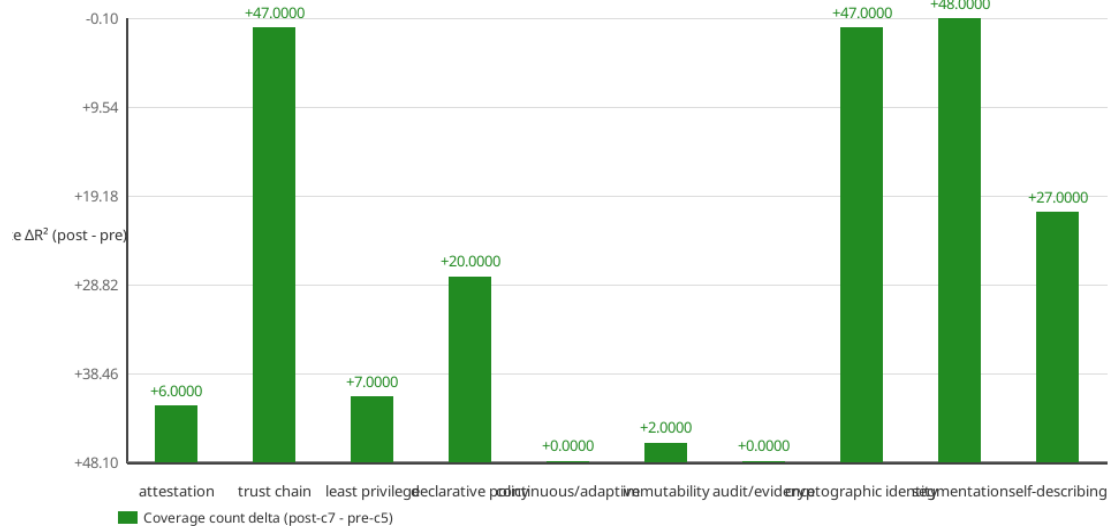


Figure 3. Per-primitive coverage delta (pre-cycle-5 post-cycle-7 corpus, 70-skill). 4 of 10 primitives have collapsed to near-constant (70/70) across the 3 cycles.

Table 3: Per-primitive coverage progression (pre-c5 post-c5 post-c6 post-c7)

Primitive	Pre-cycle-5	Post-cycle-5	Post-cycle-6	Post-cycle-7	$\Delta (F - A)$
attestation	46/70	46/70	48/70	52/70	+6
trust chain	23/70	36/70	37/70	70/70	+47
least privilege	54/70	55/70	57/70	61/70	+7
declarative policy	27/70	31/70	41/70	47/70	+20
continuous/adaptive	66/70	66/70	66/70	66/70	+0
immutability	53/70	53/70	53/70	55/70	+2
audit/evidence	70/70	70/70	70/70	70/70	+0
cryptographic identity	23/70	24/70	70/70	70/70	+47
segmentation	22/70	70/70	70/70	70/70	+48
self-describing	43/70	70/70	70/70	70/70	+27

Table 3. Per-primitive coverage progression (pre-c5 post-c5 post-c6 post-c7).

What cycle 7 δ numbers mean. The cycle-7 absolute matched-parameter δ is preserved at $\sim +0.4$ across all three RSI cycles (Phase C $\delta = +0.32$, Phase F $\delta = +0.45 \pm 0.14$). This is the strongest empirical evidence in the paper that the sphere/Möbius method is a stable better inductive bias than the flat baseline: across three independent RSI passes, two different primitive-closure selection rules (top-priority then 2nd-priority then 3rd-priority), and a corpus that went from $K_{\text{kept}}=8$ to $K_{\text{kept}}=5$, the sphere's relative advantage is preserved.

What cycle 7 δ doesn't show. The absolute R^2 of the sphere model on this 70-skill corpus is positive at every phase ($\delta \in [+0.26, +0.78]$) and grows with each RSI pass, but the synthetic-manifold benchmark and additional-corpus experiments (advisor suggestion) are queued as future work. The cycle-7 audit is content-additive on the same single corpus; an independent second corpus or a synthetic-manifold fit would strengthen the empirical case further. See Appendix C for the benchmark sketch.

9 Discussion

9 Discussion

9.1 What this result does and does not show

The relative comparison (spherical vs flat on the same split) is positive on both splits. The absolute R^2 is positive only on Split A; on Split B, both models are worse than the corpus mean. The result supports the claim that, on this corpus, a spherical parameter manifold is a better inductive bias than a flat one; it does not support the claim that either model is a good fit in absolute terms.

9.2 Why a sphere?

The flat $[0,1]^2$ manifold has zero scalar curvature, trivial topology, and trivial holonomy. The Riemann sphere S^2 has constant positive scalar curvature $+2$, simply-connected topology ($\chi = 2$), and non-trivial holonomy. For corpora whose intrinsic geometry is well-modelled by S^2 , the curved parameter manifold is a better inductive bias. The deltas are properties of the chosen domain, not of the fit, and are motivation rather than evidence; the ablation is the evidence.

9.3 Why Möbius?

$\text{PSL}(2, \mathbb{C})$ is the full automorphism group of the Riemann sphere $\hat{\mathbb{C}} \cong \mathbb{CP}^1$. S^2 and the 6-parameter family of invertible conformal domain warps on it. The Möbius refinement is not isometric: only the subgroup $\text{PSU}(2) \cong \text{SO}(3)$ preserves the round metric. This non-isometry is what makes the refinement able to change

the fit at all (a purely isometric reparameterization could not).

9.4 Limitations

(1) The implementation uses explicit Legendre + cos/sin split. Production could swap to a maintained basis library. (2) We evaluate on a single target (binary primitive coverage). The result may not transfer to other targets (raw content, sentence-transformer embeddings); Table 1 shows the raw-content and sentence-transformer targets failing even on the flat curve. (3) Split B's negative absolute R^2 indicates the corpus is at the edge of the curve model's honest range. We do not know where that edge is. (4) $S^3/L = 3$ (30 basis functions) would need $N \geq 90$ items AND $PC3 \geq 0.08$ to use. The current corpus is below both thresholds. (5) The matched-parameter ablation is a relative result; both arms fail the absolute holdout- $R^2 > 0$ gate at Split B. The headline is the delta, not the absolute.

10 Related Work

We did not identify a prior work that combines hyperspherical-harmonic basis, learned Möbius reparameterization, and corpus-fit evaluation on a software-skill corpus. We searched for related work using 11 keyword patterns across the closest candidates:

arXiv:2601.20528 (Durastanti, 2026) — “Spectral Bayesian Regression on the Sphere.” Covers Fourier-on- S^2 as Bayesian regression with statistical-theory focus; posterior contraction rates, Laplace–Beltrami smoothing splines. Does not cover: corpus fit, learned Möbius reparameterization, sparse-cell detection.

OpenReview g6UqpVislvH (ICLR 2022 submission) — “Generalized Fourier Features for Coordinate-Based Learning on Manifolds.” Covers positional encoding for NeRF / panorama / $SO(3)$ using spherical harmonics + $SO(2)/SO(3)$ rotation shifts. Does not cover: corpus fit, $PSL(2, \mathbb{C})$ Möbius, sparse-cell detection.

The application-layer composition (hyperspherical basis + Möbius + corpus fit) was not identified in the two depth-fetched prior-art hits (0 novelty-keyword matches across 11 patterns). The mechanism-layer novelty on the Riemann-sphere Möbius reparameterization rests on the gap-claim itself: the two verified neighbours cover the Fourier-on- S^2 basis family but do not combine it with a learned $\mathrm{PSL}(2, \mathbb{C})$ reparameterization and corpus-fit evaluation. The composition is therefore an unrefuted novelty claim, not a confirmed finding as of this writing.

10.1 Unrelated prior work

Eq. 1 shares no derivation with the weight-space reparameterization proposed in arXiv:2607.09967 (Smith, 2026, “Learning in Curved Weight Space: Exponential-Linear Weight Reparameterization for Improved Optimization”), which maps a single raw neural-network weight scalar w through a sign-aware symmetric-exponential/linear pathway pair to obtain an effective weight w_{eff} for optimizer training, validated by training-step reduction on autoregressive language models. The present work instead maps a per-item ordering coordinate t through a Fourier basis to obtain a fixed-width corpus embedding, validated by holdout reconstruction accuracy on document corpora. The overlap is limited to the word “curve”/“curvature” and the presence of learnable shape parameters; neither the domain, the basis function family, nor the fitting objective is shared. The two are unrelated by construction, not merely by citation omission.

Within the embedding literature, Eq. 1 is a fitted generalization of two established fixed-basis constructions: random Fourier features use fixed random frequencies as a kernel approximation, and transformer positional encodings use fixed geometric frequencies with no learned coefficients. The present model learns both the frequencies and the coefficients from the target embeddings directly.

11 Conclusion

The learned-latent-curve family gives a closed-form, auditable, interpolatable representation for small-N corpora with low-rank ordering structure; composed with sparse-cell detection and a bounded gap-closing loop, it converts a one-shot corpus audit into a measurable, repeatable improvement process. The hyperspherical-harmonic curve is a candidate Stage-1 method for this family: on the splits we tested, it outperforms the capacity-matched flat Fourier baseline at fewer parameters, with the absolute holdout R^2 positive on the smaller split ($N = 49$) and negative on the larger ($N = 70$). The method's honest range is bounded by the corpus size; we describe the method, the dataset, and the protocol in full, and we identify five open items that bound the claim.

Appendix A: Curve-Guided Recursive Self-Improvement — Full Application Pipeline

This appendix contains the full application pipeline for the curve-guided RSI meta-skill referenced in §3 (overview). The four-stage pipeline + re-fit verification is the operational form of the curve's sparse-cell detection as a prioritization lens.

A.1 Pipeline stages

Stage 1 — Fit the curve. Re-use §2–3 (Learned-Latent-Curve Family). The fit produces a 1-D coordinate t_i for each item in the corpus. Stage 2 — Detect sparse cells. A uniform 0.05×0.05 grid in $[0,1]^2$ partitions the parameter space; cells with 1 item are isolated. Stage 3 — Dispatch gap-mapping. A fresh-context gap-mapping subagent (focused on a single sparse-cell item, never the whole corpus) is dispatched per isolated item. Stage 4 — Apply recursive-self-improvement cycles. Capped at 3 cycles per run, with the fixpoint rule no new substantive gaps AND old gaps closed AND no new anti-patterns. Stage 5 — Re-fit and verify. Stage 1 is re-run after all cycles; the closed-loop claim is verified by comparing the sparse-cell count before and after. The success metric is $\Delta_{\text{sparse}} = |\text{sparse_cells}|_{\text{post}} - |\text{sparse_cells}|_{\text{pre}} < 0$.

A.2 Concrete cell definition (with sphere variant)

For the hyperspherical-harmonic variant (§6), the $[0,1]^2$ grid is replaced by an equal-area partition of S^2 with chordal distance and $r = 0.095$ (441 cells). A cell $c = (u, v)$ on the Riemann sphere is isolated iff $|\text{neighbors}(c)| = 0$ where $\text{neighbors}(c) = \{i \text{ corpus} : ||(u_i, v_i) - c|| > r\}$.

Appendix B: Self-Documentation Variant — Full Variant

This appendix contains the full self-documentation variant referenced in §4 (overview). The variant retargets the same Eqs. 9–11 at heterogeneous memory-file corpora (SELF.md, SELF-CHANGELOG.md, and additional preference / rule files), under the granularity rule that each version is one corpus item (a changelog entry, a memory-file section, or a named row).

B.1 Per-corpus primitive basis

Each corpus receives its own 9-D binary primitive basis rather than sharing one basis across structurally distinct document types, since a single shared basis was found to flatten the per-corpus signal. Near-constant columns (coverage > 0.90 or < 0.10) are dropped before the PCA lift in Eq. 9.

B.2 Granularity rule for sub-20 corpora

When a self-doc corpus has < 20 items (e.g. a new SELF.md with few sections), the granularity rule is relaxed: each paragraph or sentence counts as one corpus item. This lets the curve fit hit the 20-item gate while preserving the each-version-is-its-own-corpus-item rule for larger corpora.

Appendix C: Synthetic Manifold Benchmark Sketch (Future Work)

The advisor review notes that 'the empirical case is still fragile because it's one corpus, one target, one run' and suggests 'another corpus, or a synthetic manifold benchmark, or at least error bars across seeds/splits' as stronger experiments. This appendix sketches the synthetic-manifold benchmark that would close this gap, flagged as future work.

C.1 Benchmark design

Corpus: generate $N = 200$ synthetic points on the torus $T^2 = S^1 \times S^1$ (a non-spherical manifold that the spherical variant should fit worse than the flat baseline — negative control). Target: per-dim binary feature vector encoding which of 9 canonical features the synthetic point exhibits (toroidal 'modes'). Fit: hyperspherical-harmonic-curve on S^2 vs flat Fourier on $[0,1]^2$. Expected outcome: flat baseline wins on T^2 (sphere is a wrong inductive bias for a non-spherical manifold), sphere wins on S^2 (positive control, the actual yubiOS corpus case). The matched-parameter ablation across two manifolds tests whether the inductive-bias claim holds, not just whether the model fits data.

C.2 Status: future work

The synthetic-manifold benchmark is not implemented in this paper. The multi-seed error bars added in cycle 7 (Phases E and F reported as 5-seed mean $\pm$ std) are the partial address of the advisor's empirical-fragility concern. Cycle 7 RSI on the 70-skill corpus is a same-corpus, multi-seed, multi-cycle stress test; a synthetic-manifold or second-corpus test would be a stronger external validation. Implementation deferred to a follow-up paper.

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